NUCLEAR POWER CORPORATION OF INDIA LIMITED

WELDING PROCEDURE SPECIFICATION (WPS) - ASME QW-482

SECTION 1 - IDENTIFICATION			
WPS NO.	NPCIL-STD-SM-11	REVISION NO.	R-1
DATE	2014-07-07	QUALIFICATION STATUS	Qualified
SUPPORTING PQR NO.(S)	A		

QA APPROVED

SECTION 2 - WELDING PROCESSES			
WELDING PROCESS (ROOT)	SMAW	WELDING PROCESS (FILL)	SMAW
TYPE	Manual	JOINT DESIGN	Groove
BACKING	With or Without	BACKING MATERIAL	As per design document

SECTION 4 - BASE METALS (QW-403)			
P-NO.	P-No. 1 Group 1/2	TO P-NO.	P-No. 1 Group 1/2
MATERIAL SPEC / GRADE	IS 2062 or IS1239 or equivalent or P1		
THICKNESS RANGE (GROOVE)	1.5 mm to 19 mm	THICKNESS RANGE (FILLET)	All Thickness

SECTION 5 - POSITIONS (QW-405)			
POSITION(S) OF GROOVE	All	WELDING PROGRESSION	Vertical Uphill
POSITION(S) OF FILLET	All	PWHT	Not Required

PREPARED BY	REVIEWED BY (QA)	APPROVED BY

SECTION 6 - PREHEAT & PWHT (QW-406 / QW-407)			
PREHEAT TEMP, MIN	10.0 °C	INTERPASS TEMP, MAX	250.0 °C
PWHT REQUIREMENTS	Not Required		

SECTION 8 - ELECTRICAL CHARACTERISTICS (QW-409)	
CURRENT / POLARITY	DC-EP
TRAVEL SPEED RANGE	40-80 mm/min

SECTION 9 - TECHNIQUE (QW-410)	
STRING OR WEAVE BEAD	Root: Stringer / Fill: Weave
CLEANING METHOD	Brushing or Grinding, joint dry prior to welding
PEENING ALLOWED	False

SECTION 10 - FILLER METALS (QW-404)			
SFA SPECIFICATION	SFA 5.1	AWS CLASSIFICATION	E-7018 or E7018-1
F-NO.	2 or 4	A-NO.	1

NOTES & RESTRICTIONS:

Anchor WPS from provided pages 15-17. Qualified WPS with PQR A; user must check drawing/spec suitability and obtain document acceptance approval.

RESTRICTION: For vertical uphill progression, maximum thickness limited to 10 mm per QW-403.6 interpretation

JOINT SCHEMATIC